Electrical Instrumentation Part-II
PowerPack Phase 4
Data Converters: DAC, ADC, Codes and Errors

Reverse-engineered semester preparation notes

Diagram-safe, derivation-first, exam-answer style

Semester Command

This Phase 4 PowerPack is for mastering data converter questions. Learn every answer in this order:

$\boxed{\text{Definition} \rightarrow \text{Need} \rightarrow \text{Diagram} \rightarrow \text{Working} \rightarrow \text{Formula} \rightarrow \text{Errors} \rightarrow \text{Numerical}}$

For every diagram, use straight and perpendicular wires. Put labels far from components. If a label becomes long, split it into two or three lines inside a box.

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1. Phase 4 Target: What You Must Be Able To Answer

After completing this phase, you must be able to answer:

1. What is a DAC and why is it required?
2. Explain binary-weighted resistor DAC.
3. Explain R-2R ladder DAC considering 3 bits.
4. Derive output voltage equation of DAC.
5. Explain DAC specifications: resolution, full-scale output, LSB size, settling time.
6. Explain DAC errors: offset error, gain error, linearity error and differential non-linearity.
7. Solve DAC numerical questions with offset and full-scale values.
8. What is an ADC and why is it required?
9. Explain quantization and quantization error.
10. Explain counter-type ADC.
11. Explain successive approximation register ADC with flowchart.
12. Explain dual-slope integrating ADC.
13. Compare ADC types from exam point of view.

Must Know

Data converter questions are very scoring because they repeat the same pattern: draw the circuit or block diagram, write the conversion formula, then explain the error or numerical step.

2. Digital and Analog Signals from Zero

2.1. Analog Signal

An analog signal can take any value within a continuous range.

Example:

$$0.1\,\mathrm{V}, \quad 0.15\,\mathrm{V}, \quad 0.151\,\mathrm{V}, \quad 0.1517\,\mathrm{V}$$

Therefore,

$$\boxed{\text{Analog signal is continuous in amplitude.}}$$

2.2. Digital Signal

A digital signal takes only a finite number of discrete values.

For binary digital systems,

$$\boxed{0 \quad \text{and} \quad 1}$$

are used.

2.3. Why Converters are Needed

Most physical quantities are analog:

$$\text{temperature, pressure, speed, voltage, current, displacement}$$

But computers and digital instruments process digital data. Hence,

$$\boxed{\text{ADC converts analog to digital}}$$

and

$$\boxed{\text{DAC converts digital to analog}}$$

2.4. Converter Chain Diagram

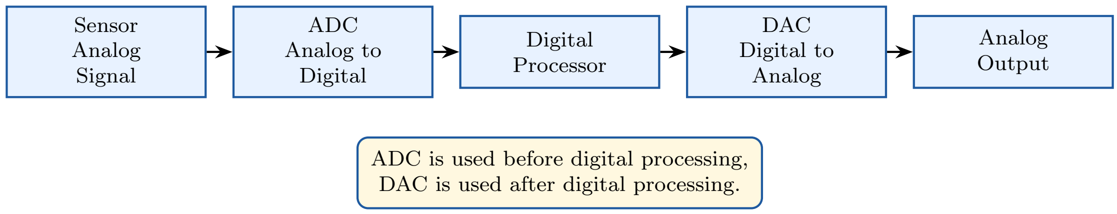

Figure 1: General signal chain using ADC and DAC.

3. DAC: Digital to Analog Converter

3.1. Definition

A Digital to Analog Converter, or DAC, is an electronic circuit which converts a digital binary input into an equivalent analog output voltage or current.

$$\boxed{\text{DAC converts binary code into proportional analog signal.}}$$

If the input is an n-bit binary number,

$$b_{n-1}b_{n-2}\cdots b_1b_0$$

then its decimal equivalent is

$$D = b_{n-1}2^{n-1} + b_{n-2}2^{n-2} + \cdots + b_1 2^1 + b_0 2^0$$

where each bit is either 0 or 1.

3.2. Ideal DAC Output

For an ideal unipolar DAC,

$$\boxed{V_o = V_{\text{LSB}} D}$$

where,

$$D = \text{decimal equivalent of digital input}$$

and,

$$V_{\text{LSB}} = \text{one step size}$$

If the full-scale reference is V_{ref}, then for an n-bit DAC:

$$\boxed{V_{\text{LSB}} = \frac{V_{\text{ref}}}{2^n}}$$

and,

$$\boxed{V_o = V_{\text{ref}}\left[\frac{b_{n-1}}{2} + \frac{b_{n-2}}{4} + \cdots + \frac{b_0}{2^n}\right]}$$

Do Not Write This Mistake

Do not write $V_{\text{LSB}} = V_{\text{ref}}/(2^n - 1)$ unless the problem specifically defines full-scale output as the output at all-ones code. In most converter formula derivations, ideal step size is taken as $V_{\text{ref}}/2^n$.

4. DAC Resolution

4.1. Definition

Resolution is the smallest change in analog output corresponding to one LSB change in digital input.

$$\boxed{\text{Resolution} = 1\,\text{LSB step size}}$$

For an n-bit DAC,

$$\boxed{\text{Resolution} = \frac{1}{2^n}}$$

As percentage,

$$\boxed{\%\,\text{Resolution} = \frac{1}{2^n} \times 100}$$

In voltage form,

$$\boxed{V_{\text{LSB}} = \frac{V_{\text{ref}}}{2^n}}$$

4.2. Example

For an 8-bit DAC,

$$2^8 = 256$$

$$\text{Resolution} = \frac{1}{256}$$

$$\%\,\text{Resolution} = \frac{1}{256} \times 100 = 0.3906\%$$

4.3. Staircase Transfer Characteristic

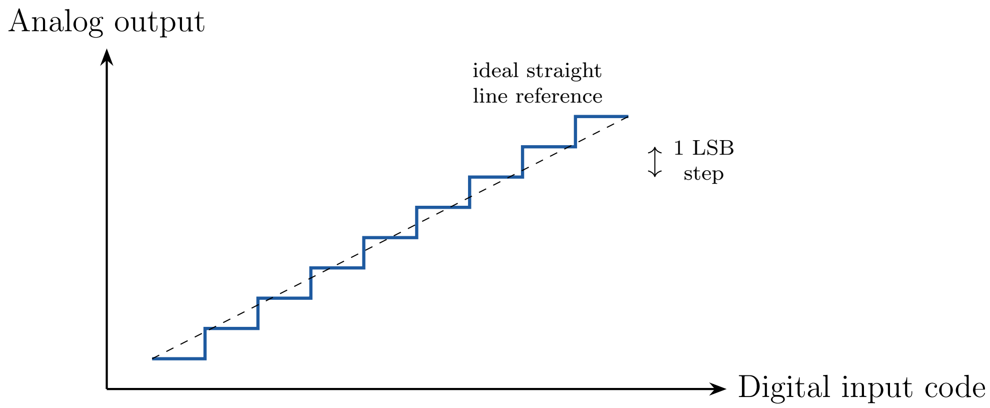

Figure 2: Ideal DAC staircase transfer characteristic.

5. Binary-Weighted Resistor DAC

5.1. Basic Idea

In a binary-weighted resistor DAC, each digital bit controls a switch connected either to V_{ref} or to ground.

The input resistors are weighted in binary order:

$$R, \quad 2R, \quad 4R, \quad 8R, \ldots$$

The MSB has the smallest resistor, so it contributes the largest current.

5.2. 3-bit Binary-Weighted DAC Diagram

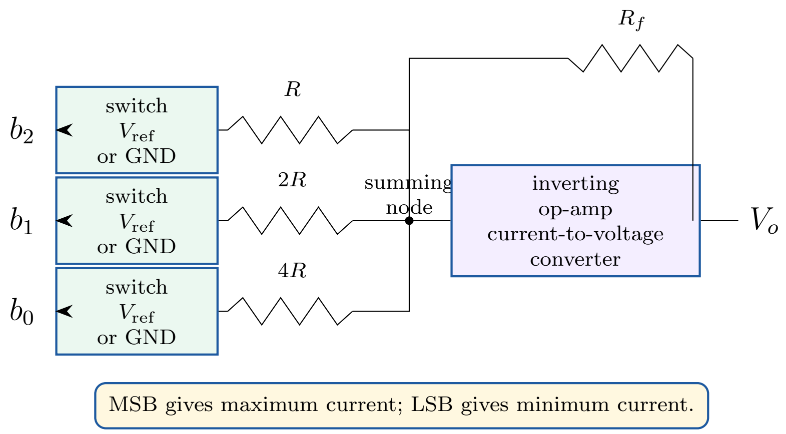

Figure 3: 3-bit binary-weighted resistor DAC.

5.3. Output Equation

For a 3-bit binary-weighted DAC,

$$I_{\text{total}} = \frac{b_2 V_{\text{ref}}}{R} + \frac{b_1 V_{\text{ref}}}{2R} + \frac{b_0 V_{\text{ref}}}{4R}$$

For an inverting op-amp,

$$V_o = -R_f I_{\text{total}}$$

Therefore,

$$\boxed{V_o = -R_f V_{\text{ref}} \left[\frac{b_2}{R} + \frac{b_1}{2R} + \frac{b_0}{4R} \right]}$$

If $R_f = R/2$, then,

$$\boxed{V_o = -V_{\text{ref}}\left[\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8}\right]}$$

6. R-2R Ladder DAC

6.1. Why R-2R Ladder DAC is Preferred

Binary-weighted DAC requires many different resistor values:

$$R, \quad 2R, \quad 4R, \quad 8R, \quad 16R, \ldots$$

For large bit numbers, this becomes difficult to fabricate accurately.

R-2R ladder DAC uses only two resistor values:

$$\boxed{R \quad \text{and} \quad 2R}$$

Therefore, it is easier to fabricate and more accurate.

6.2. 3-bit R-2R Ladder DAC Diagram

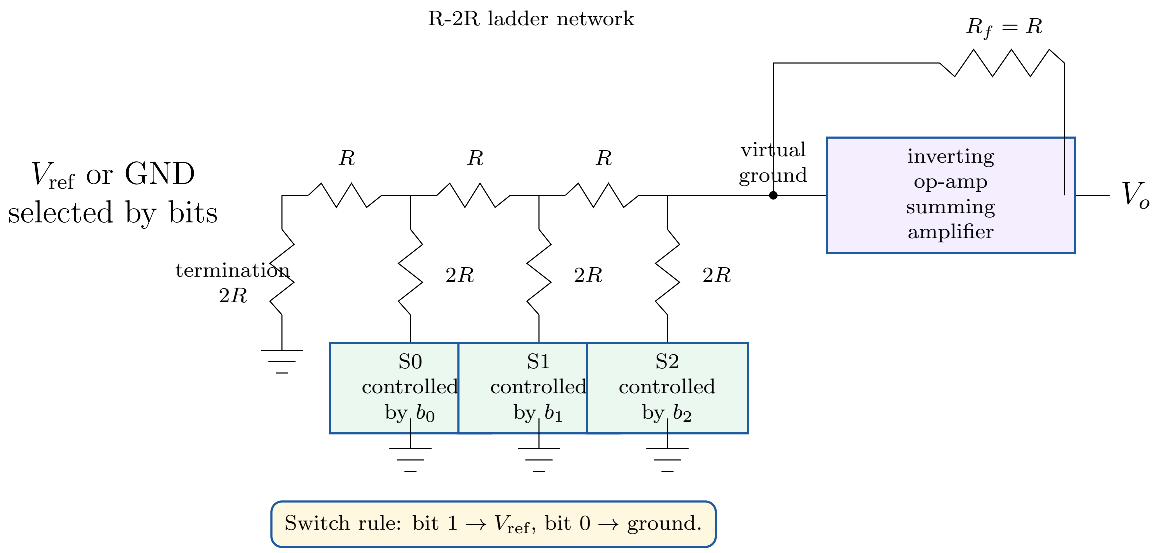

Figure 4: 3-bit R-2R ladder DAC using an inverting op-amp.

6.3. R-2R DAC Output Equation

For a 3-bit DAC,

$$b_2 = \text{MSB}, \qquad b_1 = \text{middle bit}, \qquad b_0 = \text{LSB}$$

The binary-weighted output is,

$$\boxed{V_o = -V_{\text{ref}}\left[\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8}\right]}$$

The negative sign appears because the op-amp is in inverting mode.

Magnitude:

$$\boxed{|V_o| = V_{\text{ref}} \left[\frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8} \right]}$$

7. DAC Errors

7.1. Offset Error

Offset error means the output is not zero when the input code is all zero.

For ideal DAC,

$$0000 \rightarrow 0\,\mathrm{V}$$

But in practical DAC,

$$0000 \rightarrow V_{\text{offset}}$$

Thus,

$$\boxed{\text{Offset error shifts the whole transfer curve up or down.}}$$

7.2. Gain Error

Gain error means the slope of the actual transfer characteristic is different from the ideal slope after removing offset error.

$$\boxed{\text{Gain error changes the slope.}}$$

7.3. Linearity Error

Linearity error is the maximum deviation of the actual DAC output from the ideal straight line after offset and gain errors have been removed.

$$\boxed{\text{Linearity error} = V_{\text{actual}} - V_{\text{ideal straight-line}}}$$

7.4. Differential Non-Linearity

Differential non-linearity, or DNL, is the deviation of an actual step size from the ideal 1 LSB step.

$$\boxed{\text{DNL} = \text{actual step size} - 1\,\text{LSB}}$$

If DNL is very large, the DAC may become non-monotonic.

7.5. DAC Error Diagram

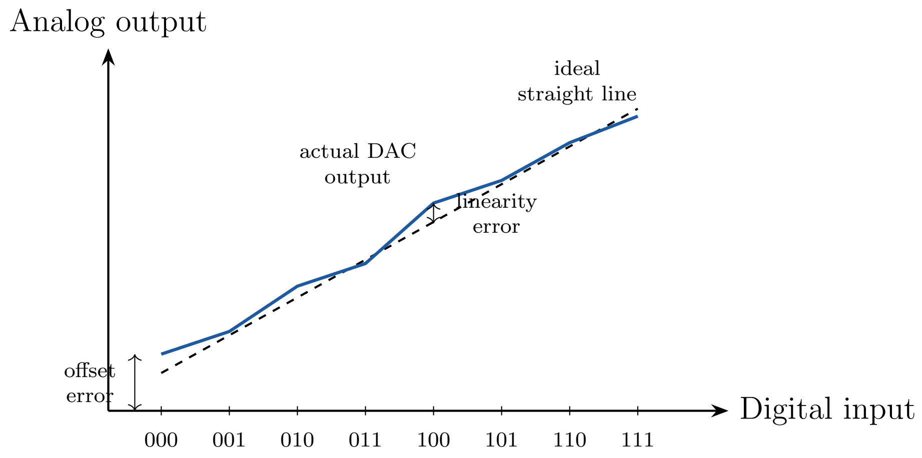

Figure 5: DAC transfer characteristic showing offset and linearity error.

8. DAC Numerical Templates

8.1. Template 1: Output from Given Step Size

If the output for code 0000 is V_{0000}, and output for 0001 is V_{0001}, then

$$\boxed{1\,\text{LSB} = V_{0001} - V_{0000}}$$

For input decimal value D,

$$\boxed{V_o(D) = V_{0000} + D(1\,\text{LSB})}$$

8.2. Example

Given,

$$V_{0000} = 0.0125\,\text{V}$$

$$V_{0001} = 0.0625\,\text{V}$$

Then,

$$1\,\text{LSB} = 0.0625 - 0.0125$$

$$\boxed{1\,\text{LSB} = 0.05\,\text{V}}$$

For input 1110_2,

$$1110_2 = 8 + 4 + 2 + 0 = 14$$

Thus,

$$V_o = 0.0125 + 14(0.05)$$

$$V_o = 0.0125 + 0.7000$$

$$\boxed{V_o = 0.7125\,\text{V}}$$

8.3. Template 2: Offset Error in DAC

If offset error is given in LSB,

$$\boxed{V_{\text{actual}} = V_{\text{ideal}} + V_{\text{offset}}}$$

For offset error of -0.5 LSB,

$$\boxed{V_{\text{actual}} = V_{\text{ideal}} - 0.5\,\text{LSB}}$$

9. ADC: Analog to Digital Converter

9.1. Definition

An Analog to Digital Converter, or ADC, is an electronic circuit which converts an analog input voltage into an equivalent digital binary output.

$$\boxed{\text{ADC converts analog signal into digital code.}}$$

9.2. Basic ADC Operation

ADC operation has three conceptual steps:

$$\boxed{\text{Sampling} \rightarrow \text{Quantization} \rightarrow \text{Encoding}}$$

1. **Sampling:** continuous-time signal is observed at discrete instants.
2. **Quantization:** sampled amplitude is approximated to nearest allowed level.
3. **Encoding:** quantized level is represented by binary code.

9.3. ADC Block Diagram

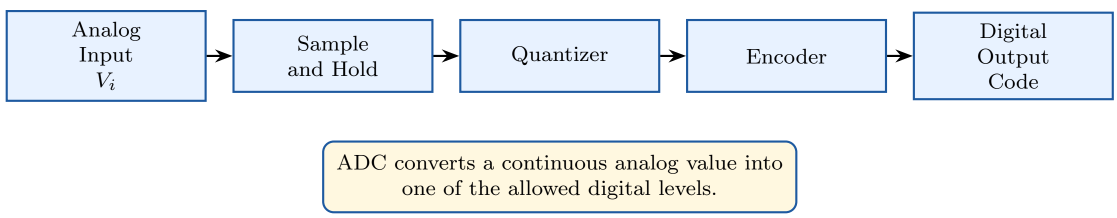

Figure 6: Basic operation chain of an ADC.

10. Quantization and Quantization Error

10.1. Quantization

Quantization is the process of approximating a continuous analog amplitude to the nearest discrete digital level.

For an n-bit ADC,

$$\boxed{\text{number of levels} = 2^n}$$

If the full-scale input range is V_{FS}, then

$$\boxed{1\,\text{LSB} = \frac{V_{FS}}{2^n}}$$

10.2. Quantization Error

Since the analog value is rounded to the nearest digital level, a small error occurs. This is called quantization error.

Maximum quantization error is

$$\boxed{\pm\frac{1}{2}\text{LSB}}$$

10.3. Quantization Diagram

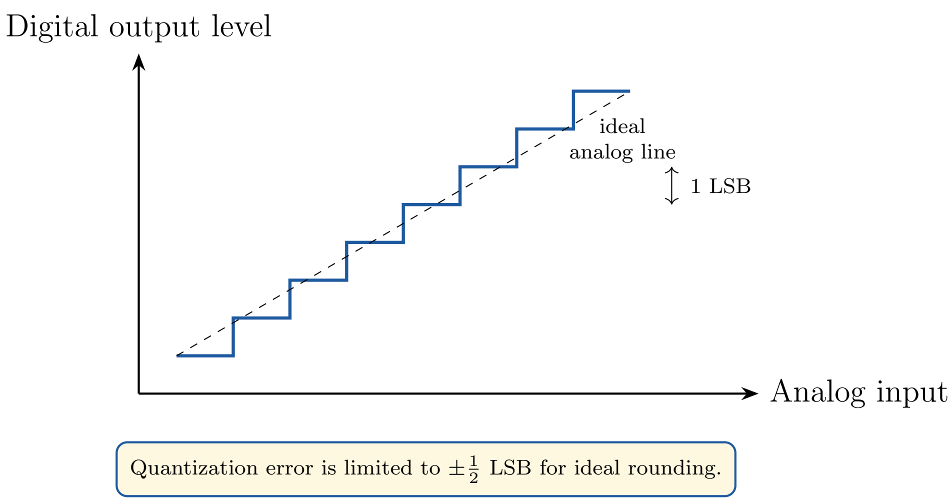

Figure 7: Quantization staircase and LSB step.

11. Counter-Type ADC

11.1. Principle

A counter-type ADC uses a counter, DAC and comparator.

The counter starts from zero and increases step by step. The digital count is converted into an analog voltage by the DAC. This DAC output is compared with the input analog voltage.

When,

$$V_{\text{DAC}} \geq V_i$$

the counter stops. The counter output is the digital equivalent.

11.2. Counter-Type ADC Block Diagram

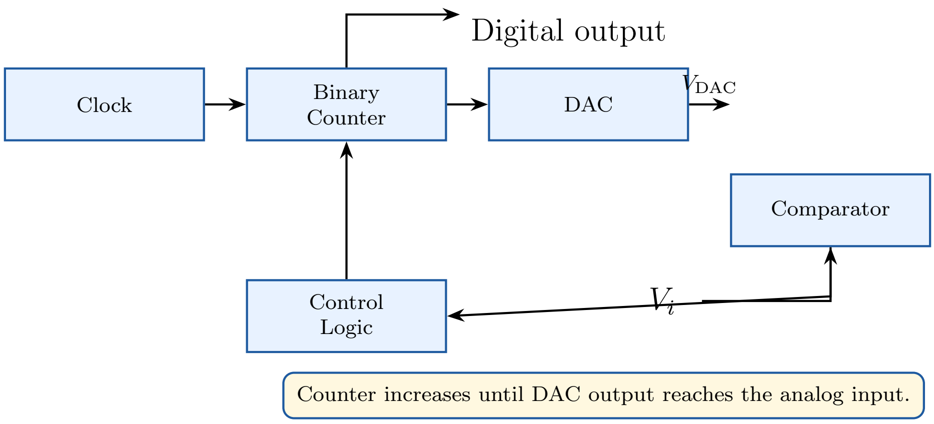

Figure 8: Counter-type ADC block diagram.

11.3. Advantages and Disadvantages

Advantages	**Disadvantages**
Simple circuit	Slow conversion speed
Easy to understand	Conversion time depends on input voltage
Uses counter and DAC	Worst case requires $2^n - 1$ clock pulses

12. Successive Approximation Register ADC

12.1. Principle

A Successive Approximation Register ADC, or SAR ADC, finds the digital output by testing one bit at a time, starting from the MSB.

It is faster than counter-type ADC because it does not count from zero to the final value. It performs a binary search.

For an n-bit SAR ADC,

$$\boxed{\text{conversion needs } n \text{ clock cycles}}$$

12.2. SAR ADC Block Diagram

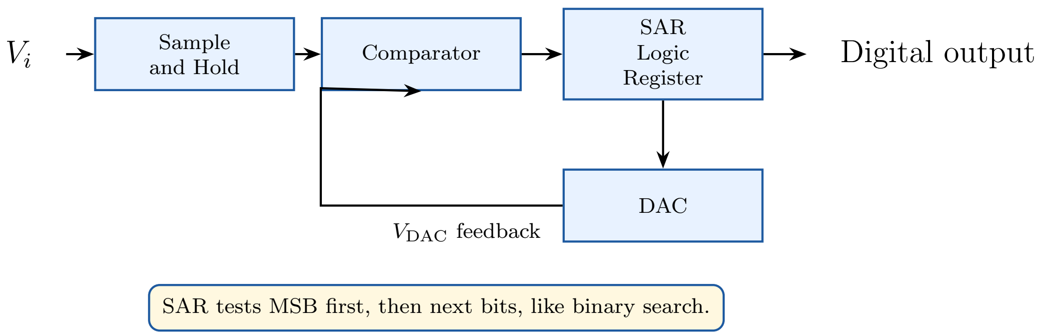

Figure 9: Successive approximation register ADC block diagram.

12.3. Working of SAR ADC

Step 1: The sample and hold circuit holds the input voltage constant during conversion.

Step 2: SAR sets the MSB to 1 and all other bits to 0.

Step 3: DAC converts this trial code into analog voltage V_{DAC}.

Step 4: Comparator compares V_i and V_{DAC}.

Step 5: If $V_i \geq V_{\text{DAC}}$, the trial bit is kept as 1.

Step 6: If $V_i < V_{\text{DAC}}$, the trial bit is reset to 0.

Step 7: The same process continues for the next lower bit.

Step 8: After n trials, the SAR output is the final digital equivalent.

12.4. SAR Flowchart

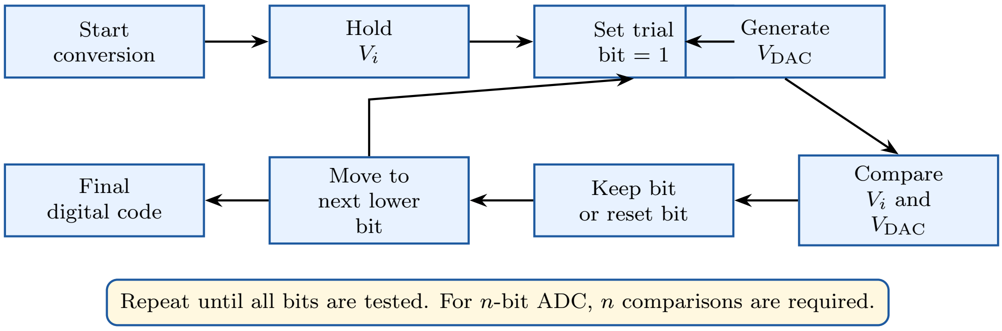

Figure 10: Flowchart of SAR ADC operation.

13. Dual-Slope Integrating ADC

13.1. Principle

A dual-slope ADC converts analog voltage into time. It uses an integrator, comparator, counter and clock.

It has two phases:

$$\boxed{\text{Phase 1: integrate unknown input}}$$

$$\boxed{\text{Phase 2: integrate reference voltage of opposite polarity}}$$

The time required in the second phase is proportional to the input voltage.

13.2. Dual-Slope ADC Block Diagram

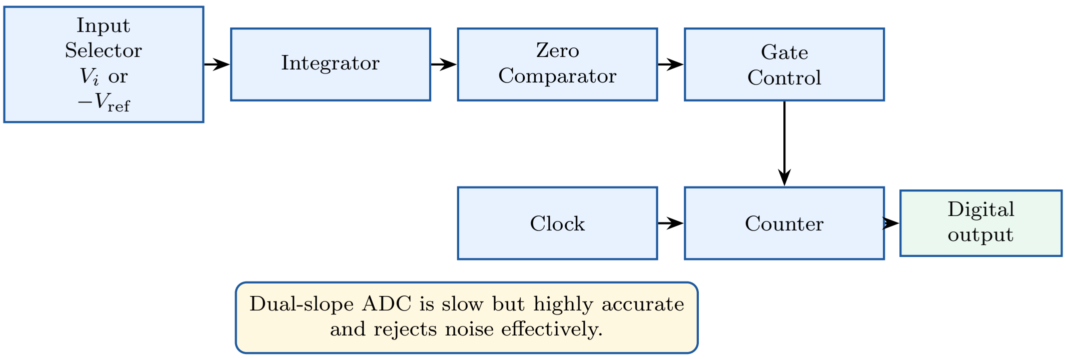

Figure 11: Dual-slope integrating ADC block diagram.

13.3. Derivation

During fixed integration time T_1, the input voltage V_i is applied to the integrator.

Integrator output after time T_1:

$$V_o(T_1) = -\frac{1}{RC}\int_0^{T_1} V_i\,dt$$

Since V_i is constant,

$$V_o(T_1) = -\frac{V_i T_1}{RC}$$

During the second phase, a reference voltage of opposite polarity is applied. The output returns to zero in time T_2.

The change during second phase is,

$$\frac{V_{\text{ref}}T_2}{RC}$$

For returning to zero,

$$\frac{V_iT_1}{RC} = \frac{V_{\text{ref}}T_2}{RC}$$

Cancel RC:

$$V_iT_1 = V_{\text{ref}}T_2$$

Therefore,

$$\boxed{T_2 = \frac{V_i}{V_{\text{ref}}}T_1}$$

Since clock pulses are counted during T_2,

$$\boxed{N = f_{\text{clk}}T_2}$$

Thus,

$$\boxed{N = f_{\text{clk}}\frac{V_i}{V_{\text{ref}}}T_1}$$

So digital count is proportional to input voltage.

$$\boxed{N \propto V_i}$$

13.4. Dual-Slope Timing Diagram

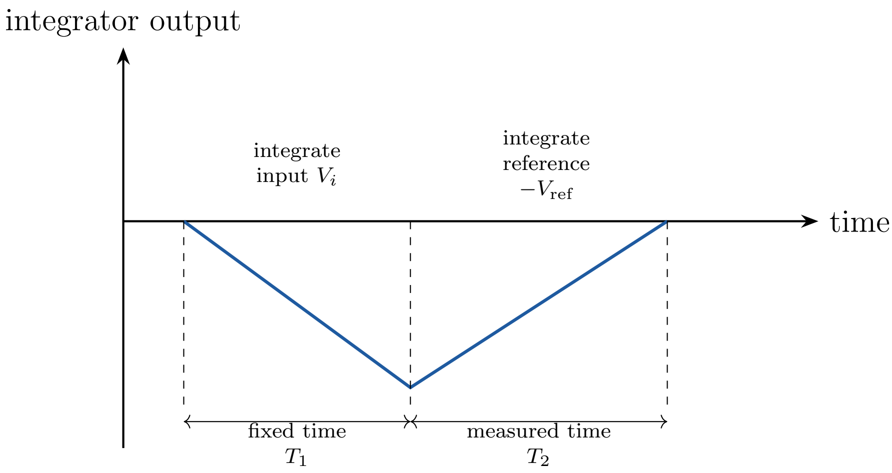

Figure 12: Integrator output waveform in dual-slope ADC.

14. ADC Numerical Template

14.1. Digital Equivalent

For an n-bit ADC with full-scale voltage V_{FS},

$$1\,\text{LSB} = \frac{V_{FS}}{2^n}$$

For input voltage V_i,

$$\boxed{D = \frac{V_i}{1\,\text{LSB}}}$$

or,

$$\boxed{D = \frac{V_i}{V_{FS}} 2^n}$$

The digital output is the binary equivalent of D, usually rounded or truncated depending on the ADC convention.

14.2. Conversion Time

For counter-type ADC,

$$\boxed{T_C = N T_{\text{clk}}}$$

Worst-case count for n-bit ADC:

$$N_{\max} = 2^n - 1$$

So,

$$\boxed{T_{C,\max} = (2^n - 1) T_{\text{clk}}}$$

For SAR ADC,

$$\boxed{T_C = n T_{\text{clk}}}$$

Must Know

Counter ADC conversion time depends on input voltage. SAR ADC conversion time is fixed for a given number of bits.

15. Comparison of ADC Types

ADC Type	**Speed**	**Accuracy**	**Important Point**
Counter type	Slow	Moderate	Counts from zero upward
SAR type	Fast	Good	Binary search; n clock cycles
Dual-slope	Slow	Very high	Excellent noise rejection
Flash ADC	Very fast	Moderate	Uses many comparators

15.1. Which ADC to Choose

1. For digital voltmeter: dual-slope ADC is preferred.

2. For medium speed instrumentation: SAR ADC is preferred.

3. For very high-speed applications: flash ADC is preferred.

4. For simple low-cost conversion: counter-type ADC can be used.

16. ADC and DAC Code Table

For a 3-bit unipolar converter:

$$2^3 = 8$$

$$1\,\mathrm{LSB} = \frac{V_{\mathrm{ref}}}{8}$$

Binary Code	Decimal D	Ideal Analog Level
000	0	0
001	1	$\frac{V_{\mathrm{ref}}}{8}$
010	2	$\frac{2V_{\mathrm{ref}}}{8}$
011	3	$\frac{3V_{\mathrm{ref}}}{8}$
100	4	$\frac{4V_{\mathrm{ref}}}{8}$
101	5	$\frac{5V_{\mathrm{ref}}}{8}$
110	6	$\frac{6V_{\mathrm{ref}}}{8}$
111	7	$\frac{7V_{\mathrm{ref}}}{8}$

Semester Command

Whenever a DAC or ADC numerical looks confusing, first convert the binary code into decimal. Then multiply by 1 LSB.

17. How to Recognise Data Converter Questions

Question Language	What You Must Write
"R-2R ladder DAC considering 3 bits"	Draw 3-bit R-2R ladder, explain switch rule, derive weighted output equation.
"Linearity error of DAC"	Define deviation from ideal straight line after offset and gain error removal; draw transfer curve.
"DAC output for input code"	Find LSB, convert binary to decimal, use $V_o = V_{0000} + D(1\,\text{LSB})$.
"SAR ADC with flowchart"	Draw blocks: sample-hold, comparator, SAR, DAC; explain MSB-first binary search.
"Dual-slope ADC"	Draw integrator-comparator-counter block diagram and derive $T_2 = (V_i/V_{ref})T_1$.
"ADC codes and errors"	Write quantization, $1\,\text{LSB} = V_{FS}/2^n$, quantization error = $\pm 1/2$ LSB.

18. One-Page Formula Bank

$$\boxed{D = b_{n-1}2^{n-1} + b_{n-2}2^{n-2} + \cdots + b_0}$$

$$\boxed{V_{\text{LSB}} = \frac{V_{\text{ref}}}{2^n}}$$

$$\boxed{V_o = V_{\text{LSB}} D}$$

$$\boxed{V_o = V_{\text{ref}} \left[\frac{b_{n-1}}{2} + \frac{b_{n-2}}{4} + \cdots + \frac{b_0}{2^n} \right]}$$

$$\boxed{\text{Resolution} = \frac{1}{2^n}}$$

$$\boxed{\%\,\text{Resolution} = \frac{1}{2^n} \times 100}$$

$$\boxed{\text{Quantization error} = \pm\frac{1}{2}\text{LSB}}$$

$$\boxed{D = \frac{V_i}{V_{FS}} 2^n}$$

$$\boxed{T_{C,\max} = (2^n - 1)T_{\text{clk}} \quad \text{for counter ADC}}$$

$$\boxed{T_C = nT_{\text{clk}} \quad \text{for SAR ADC}}$$

$$\boxed{T_2 = \frac{V_i}{V_{\text{ref}}} T_1 \quad \text{for dual-slope ADC}}$$

19. Final Rapid Revision for Phase 4

19.1. If Only 30 Minutes Are Left

Revise in this order:

1. DAC definition and ideal output formula.

2. R-2R ladder DAC diagram and output equation.

3. DAC errors: offset, gain, linearity, DNL.

4. ADC definition and quantization error.

5. SAR ADC block diagram and MSB-first process.

6. Dual-slope ADC derivation.

7. Numerical formulas: LSB, binary-to-decimal, conversion time.

19.2. Most Probable Diagram Commands

1. 3-bit R-2R ladder DAC.

2. DAC transfer characteristic with linearity error.

3. ADC basic block diagram.

4. Counter-type ADC block diagram.

5. SAR ADC block diagram and flowchart.

6. Dual-slope ADC block diagram and timing waveform.

Do Not Write This Mistake

Do not write only theory in data converter answers. Always include either a circuit diagram, block diagram, flowchart or transfer characteristic according to the question.

Phase 4 Completed: DAC and ADC Mastery